Sahaj Bhadja

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Summary

Full-stack software engineer pursuing an MSCS at USC with 2+ years shipping React/TypeScript interfaces, Python APIs, LLM-assisted document workflows, and Postgres-backed systems. Strong in rapid iteration, healthcare-style data validation, automation, and translating messy operations into reliable software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized SQL-backed workflow tools supporting 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexed access patterns, and service fixes.
- Built validation pipelines processing 5,000+ monthly jobs with reconciliation checks, CI/CD releases, and runbooks, improving reliability for structured reporting and operational handoffs.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React and Flask product features handling 10,000+ daily requests, converting ambiguous user needs into responsive interfaces, REST APIs, tests, and release-ready increments.
- Built AI-assisted document workflows with OCR, LLM prompts, and AWS-backed persistence, reducing feedback turnaround by **60%** while documenting extraction edge cases and failures.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python evaluation pipelines across 100,000+ records, producing reproducible reports, debugging artifacts, and visual summaries that clarified model behavior and data quality.
- Analyzed failure patterns across generated outputs and datasets, translating complex findings into actionable recommendations while strengthening validation coverage and technical decision-making.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and accelerating defect isolation across services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Frontend: React, TypeScript, JavaScript, Apollo GraphQL familiarity, responsive UI, workflow-driven interfaces
Backend: Python, Flask, FastAPI, Node.js, REST APIs, GraphQL familiarity, background jobs, business-rule automation
AI & Data: LLM prompts, document extraction, OCR, RAG, PostgreSQL, MySQL, SQL Server, MongoDB, data validation
Delivery: AWS, GCP, Docker, Git/GitHub, Jenkins, CI/CD, Airflow familiarity, testing, customer-driven iteration

Projects

HMIS Code | *MERN, TypeScript, ABDM APIs, Sentry*
- Built a healthcare management platform integrating patient, provider, and claim workflows with secure APIs, structured data validation, monitoring, and operational reliability.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a secure document Q&A platform with semantic search, summarization, and prompt workflows over 50K+ records, reducing review time by **60%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, combining schema inspection, query generation, Dockerized services, and debugging flows for faster data validation.